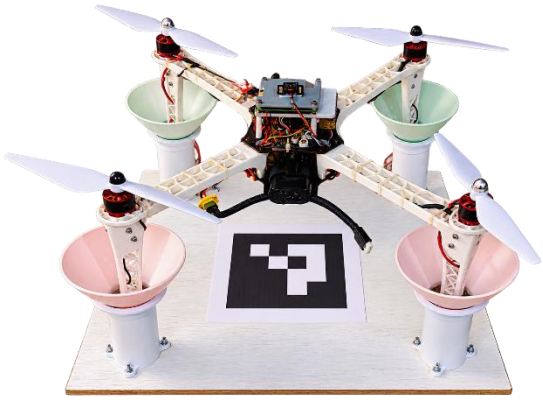
QDA-2, Autonomous Drone and GCS for Search and Mapping

Team Qbit Dynamics

Methodology

QDA-2, developed by Qbit Dynamics, integrates autonomous flight, GPS-denied localization, onboard AI perception, systematic area coverage, and autonomous docking into a unified aerial platform.

The operator defines a search area through the ground station. The UAV autonomously takes off and follows a systematic survey pattern while capturing imagery and tracking searched regions. Visual-inertial odometry (VIO), using a global-shutter camera, dedicated IMU, and laser altimeter, enables onboard position estimation without reliance on GPS. A Raspberry Pi 5 processes navigation, mission logic, and AI-based detections, while an independent STM32H7 flight controller manages flight stabilization and failsafes.



Telemetry and detection information are transmitted to the ground station, where captured imagery can be processed into stitched, georeferenced maps. Upon mission completion or reaching a low-battery threshold, the UAV returns to its docking station, lands using a vision-guided approach and passive mechanical alignment, and recharges through electrical contacts for subsequent sorties.

Proposed Outcome

The proposed outcome is a reusable, cost-conscious autonomous aerial platform for systematic surveillance, search, and mapping across diverse operational environments. Target users include border security forces, forest departments, wildlife conservation and anti-poaching teams, disaster management authorities, search-and-rescue organizations, police and public safety agencies, and infrastructure inspection teams.

Intended applications include border monitoring, wildlife surveillance, disaster assessment, missing-person searches, perimeter security, and remote-area inspection. The modular architecture allows mission-specific detection models and workflows to be developed around a common hardware platform.

GPS-denied localization, onboard processing, coverage tracking, independent flight-control failsafes, and autonomous recharging aim to improve operational continuity, reduce repetitive manual effort, and limit human exposure to hazardous or inaccessible areas. Application-specific performance will require appropriate validation and testing.

